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## **Dissolved CO<sub>2</sub> Storage in Shallow Clastic Reservoirs of Rift Basins: Example Study in Southern Red Sea**

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### **Summary**

Coastal regions, including the western area of Saudi Arabia, present opportunities for implementing effective carbon storage solutions, which can help mitigate CO<sub>2</sub> emissions. The onshore basins located at proximity to CO<sub>2</sub> sources are mostly small half-grabens formed during the opening of the Red Sea. To increase the carbon storage potential in these basins, a novel approach called dissolved CO<sub>2</sub> storage has been explored, where CO<sub>2</sub> and water are co-injected into shallow saline aquifers, allowing the CO<sub>2</sub> to rapidly dissolve and become permanently sequestered. This example study in an onshore basin of the southern Red Sea demonstrated the potential of this method, with results showing that shallow clastic saline aquifers can store approximately 6 Mt of CO<sub>2</sub> per km<sup>3</sup> of rock. The study highlights the importance of factors such as aquifer salinity, mineralogy, and injection water chemistry in determining the success of dissolved CO<sub>2</sub> storage, and contributes to the development of effective carbon management strategies in geologically complex environments, ultimately supporting the advancement of secure and sustainable CO<sub>2</sub> storage solutions.

### **Introduction**

The onshore rift basins along the Red Sea margin in Saudi Arabia offer opportunities for implementing effective carbon storage solutions to address CO<sub>2</sub> emissions (Fig. 1). These basins are mostly small half-grabens filled by syn-to post-rift siliciclastics with minor evaporites and/or volcanics. This study explores the feasibility of an innovative approach: dissolved CO<sub>2</sub> storage in shallow unconfined clastic reservoirs within rift basin settings. By co-injecting CO<sub>2</sub> and water into shallow saline aquifers (Addassi et al., 2024), the CO<sub>2</sub> rapidly dissolves in water forming a dense, CO<sub>2</sub>-saturated solution that migrates downward into the reservoir, where the CO<sub>2</sub> becomes permanently sequestered through solubility trapping. This method eliminates the need for a caprock and mitigates risks associated with dry-out and salt precipitation. This

study presents a case study from an onshore basin of southern Red Sea to demonstrate the potential of dissolved CO<sub>2</sub> storage as a viable solution for CCS deployment in challenging geological settings.

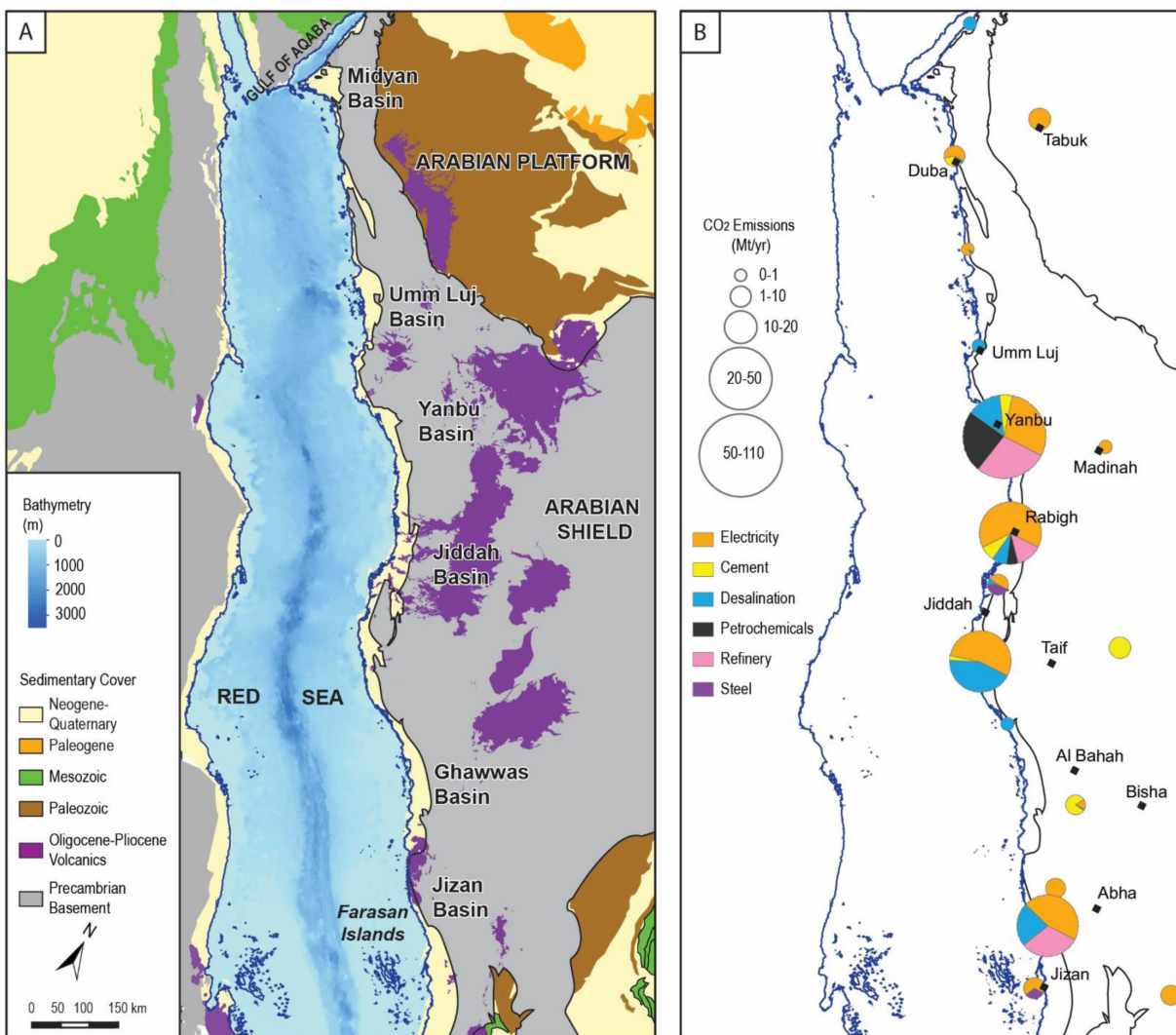


Figure 1—(a) Geological map of western Saudi Arabia (modified after Le Nindre et al., 2003) and GEBCO, 2022); (b) Map showing nearby major stationary industrial CO<sub>2</sub> emissions (modified after Hamieh et al., 2022).

Carbon capture and storage (CCS) is a critical technology in mitigating climate change by reducing CO<sub>2</sub> emissions. Deep saline aquifers, depleted oil and gas fields, and unmineable coal seams are the primary targets for the geological storage of CO<sub>2</sub>. Sequestration processes involve different trapping mechanisms depending on the hydrodynamic, physical, and chemical conditions of the reservoir. These mechanisms are typically classified into four categories (IPCC, 2005; Zhang and Song, 2014):

- Hydrodynamic trapping – CO<sub>2</sub> remains mobile and is physically confined by geological structures.
- Residual trapping – CO<sub>2</sub> becomes immobile as small bubbles trapped in pore spaces.
- Solubility trapping – CO<sub>2</sub> dissolves into formation water, reducing the risk of buoyancy-driven migration.
- Mineral trapping – Dissolved CO<sub>2</sub> reacts with native minerals to form stable carbonate precipitates over geological timescales.

Depending on the reservoir pressure and temperature, CO<sub>2</sub> may be stored as a compressed gas, liquid, or supercritical fluid. However, in most cases, a significant portion of the injected CO<sub>2</sub> remains in the mobile phase, free to migrate laterally or vertically toward the caprock. While trapping mechanisms eventually immobilize CO<sub>2</sub>, these processes take time, and there is always a risk of leakage due to caprock failure, fault activation, or pressure buildup (Kholotsa, 2022).

In the western area of Saudi Arabia, CO<sub>2</sub> emissions are estimated to be ~180 Mt/yr, accounting for ~40% country's total emissions (Fig. 1, Hamieh et al., 2022). This decarbonization challenge necessitates innovative effective carbon storage solutions to be deployed in nearby basins. Dissolved CO<sub>2</sub> storage presents a novel solution that enlarges viable carbon sinks by rendering previously inaccessible regions suitable for carbon sequestration. This method involves dissolving CO<sub>2</sub> in water before injection into shallow reservoirs, ensuring that CO<sub>2</sub> remains in the aqueous phase rather than as a free gas or supercritical fluid (Fig. 2). The technical feasibility of this solution is validated by the Carbfix project (<https://www.carbfix.com/>), which has stored more than 107 kt of CO<sub>2</sub> since 2014. However, the reservoir of Carbfix is basaltic aquifers, this scheme has not been implemented in sedimentary aquifers. The proposed scheme involves surface dissolution or in-well mixing of CO<sub>2</sub> and water (cheaper). The water supply may be from produced water or seawater.

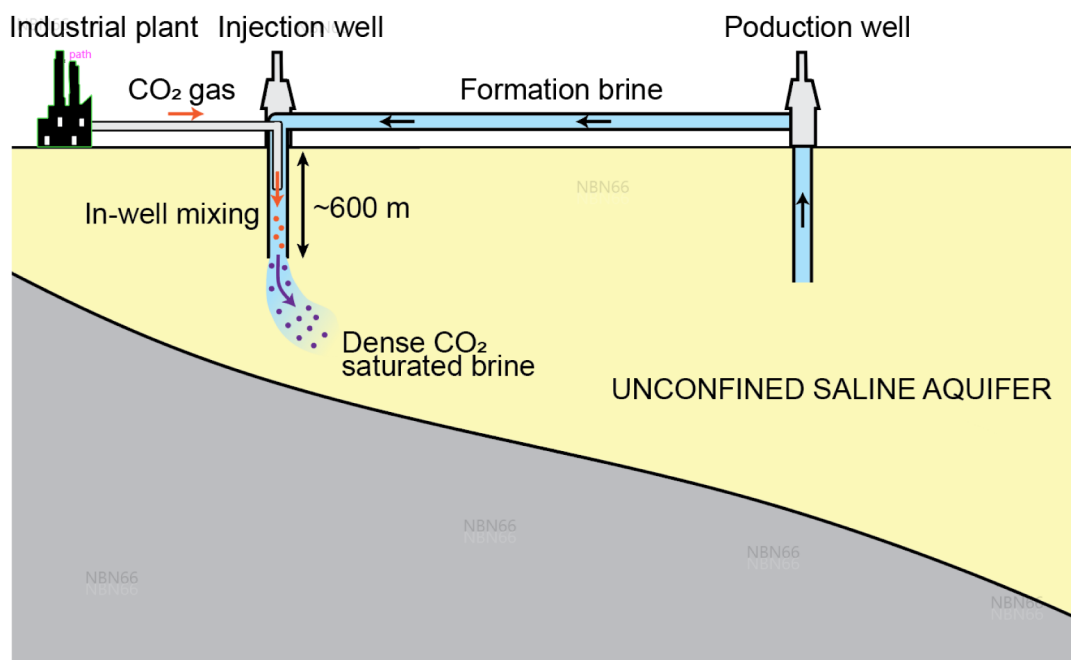


Figure 2—Schematic diagram showing the concept of dissolved CO<sub>2</sub> Storage in shallow unconfined sedimentary aquifers.

Given the potential advantages of dissolved CO<sub>2</sub> storage, this study aims to evaluate its feasibility in an onshore basin of southern Red Sea, where shallow sedimentary reservoirs provide a promising environment for its implementation. The study includes:

- 2D transport modeling of CO<sub>2</sub> injection and storage over a 100-year timeframe under different total dissolved solids (TDS) formation water and injection water.
- Analysis of CO<sub>2</sub> trapping mechanisms, including solubility trapping (HCO<sub>3</sub><sup>-</sup> formation) and mineral trapping (carbonate precipitation).
- Impact assessment of reservoir heterogeneity and formation water salinity on CO<sub>2</sub> migration and storage efficiency.

By examining these factors, this study aims to provide insights into the potential of dissolved CO<sub>2</sub> storage as a viable carbon sequestration strategy for the Red Sea coast and other coastal regions.

## Geological Settings of the Study Area

An onshore basin, located along the southern Red Sea coast, was chosen as the study area due to its favorable conditions for dissolved CO<sub>2</sub> storage and proximity to a CO<sub>2</sub> emission cluster (Fig. 1). The basin's geological history began with its formation as a large half-graben during the Oligocene opening of the Red Sea. Initially, the basin was filled with syn-rift volcanics, which were subsequently overlain by syn-rift clastics (Fig. 3). The post-rift succession consists of Miocene clastics and evaporites, as well as the Miocene-Pliocene clastics.

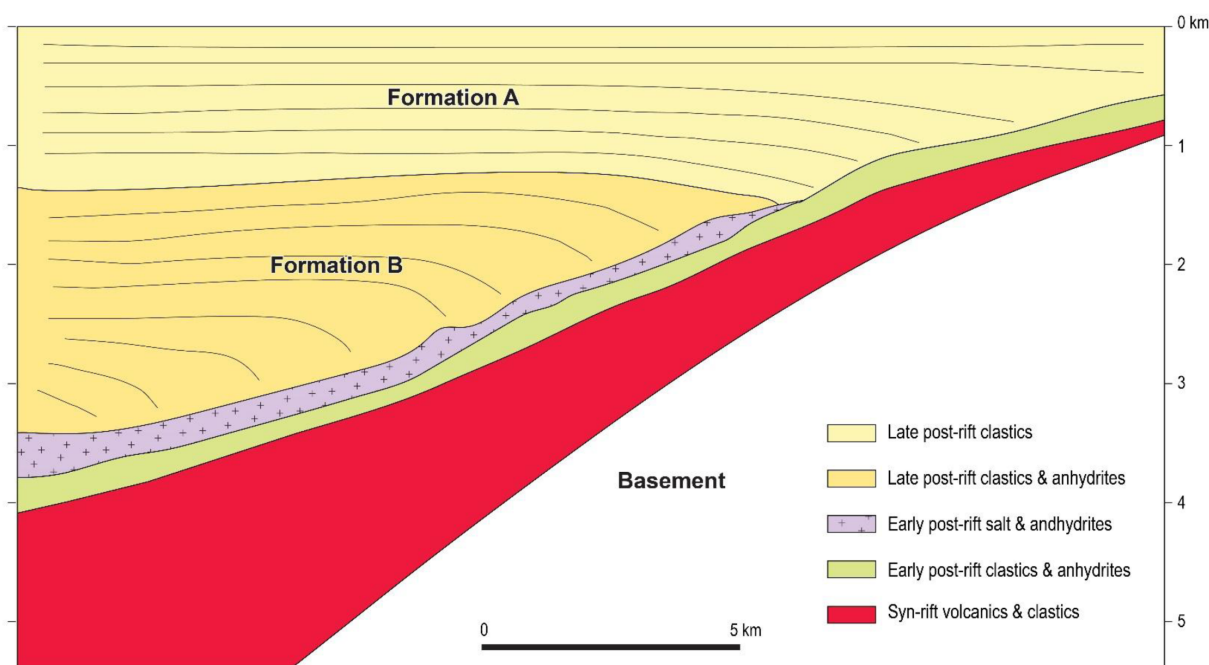


Figure 3—Schematic cross-section showing the structure and stratigraphy of the study area.

The Formations A and B characterized by roll-over structures, which formed as a result of post-depositional halokinesis of salt (Fig.3). Consequently, the thickness of these formations varies significantly across the basin, and the dip of the strata increases with depth, reaching up to approximately 20°.

The shallow Formations A and B comprise porous and permeable sandstones that are particularly suitable for dissolved CO<sub>2</sub> injection. Formation A, the shallowest of the two, consists primarily of continental to shallow marine clastics sourced from the erosion of the Arabian Shield in the east. The Upper Formation B, underlying Formation A, is composed of coastal plain sandstones and shales.

The water salinity of the Formations A and B aquifers increases rapidly with depth, reaching up to 300,000 ppm. The temperature gradient in the basin is estimated to be 42°C/km, while the pressure gradient is approximately 0.45 psi/ft for the Formations A and B. These characteristics make these clastic aquifers suitable for dissolved CO<sub>2</sub> sequestration.

## Methodology of CO<sub>2</sub> Reactive Transport Modeling

A 2D radially symmetric numerical model was developed to simulate CO<sub>2</sub> injection and storage behavior (Fig. 4). A numerical simulation program was used to simulate complex, chemically reactive flows of multi-phase fluids in porous and fractured media. The program can handle a wide range of equilibrium chemical

reactions, including aqueous complexation, gas dissolution and exsolution, and cation exchange, making it a valuable tool for modeling non-isothermal and density-dependent systems (Xu et al., 2004). In this study, the 2D models accounted for:

- Reservoir heterogeneity, incorporating variations in porosity, permeability, and mineralogical composition.
- Injection well design, assuming a single well with a perforated section of 100 m, and an injection rate of 75 kg/s ( $\sim 0.12$  Mt CO<sub>2</sub> per year per well; Fig. 5).
- Three reservoir scenarios were categorized by total dissolved solids (TDS, Scenario 1: High TDS ( $\sim 300,000$  ppm); Scenario 2: Medium TDS (270,000–290,000 ppm); Scenario 3: Low TDS (70,000–90,000 ppm) and seawater to analyze CO<sub>2</sub> transport, dissolution, and mineralization over a 100-year timescale.

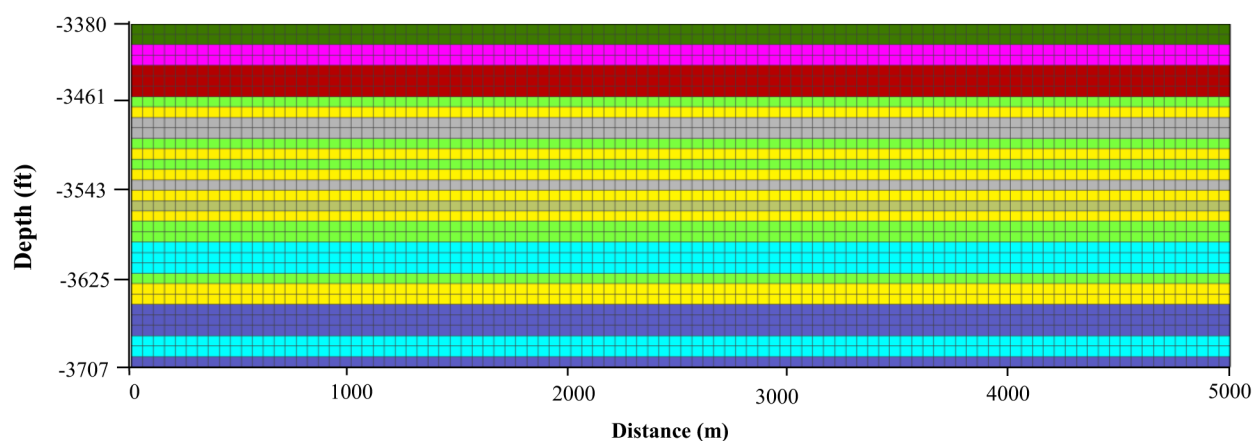


Figure 4—2D grid design for the reactive transport modeling. The color schemes are rock types with different mineralogy, porosity and permeability.

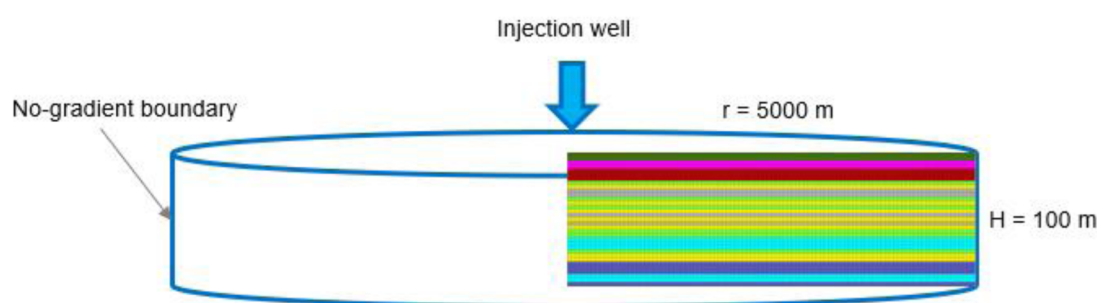


Figure 5—2D radially symmetric grid model.

The simulated CO<sub>2</sub> injection and migration patterns were used to assess the distribution of dissolved CO<sub>2</sub> and the associated chemical interactions within the reservoir.

### Initial conditions

The clastic formation is represented with 9 different rock types, assigned with different colors (Fig. 4). Generally, the lighter colors represent higher porosity and permeability and darker colors means lower porosity and permeability. The initial minerals include quartz (ranging from 15-86%), K-feldspar (ranging from 0-12.7%), oligoclase (ranging from 3.5-17.3%), illite/smectite (ranging from 0-25%), kaolinite (ranging from 0-9%), chlorite (ranging from 1.2-24%), dolomite (ranging from 0.4-12.6%), and calcite (ranging from 0-7.8%).

## Boundary conditions

In our model, a Dirichlet boundary condition was applied. This was achieved by assigning an extremely large volume, approximately  $1e50$  times that of a standard cell volume, to the boundary grid block, as described by Xu et al. (2004). As a result, the impact of influx on the boundary becomes insignificant compared to the vast volume of the formation brine. Consequently, the aqueous chemical composition and thermodynamic properties, including temperature and pressure, of the boundary element remain essentially unchanged from the original formation brine conditions, consistent with the approach outlined by Xu et al. (2004).

## Thermodynamic and kinetic data

The equilibrium constants for aqueous species and minerals used in our simulations are sourced from Xu et al. (2007). Additionally, we assume that local equilibrium is achieved among the aqueous species, implying that all reactions involving these species reach a state of equilibrium within the system.

The software uses the rate law from Lasaga et al. (1994) to calculate kinetically controlled mineral dissolution and precipitation,

$$\frac{r}{S} = \pm k a_{H^+}^i (f(\Delta G_r)) \quad (1)$$

where,  $r$  denotes the dissolution/precipitation rate in mol/s/kgw,  $S$  is the reactive surface area per kg  $H_2O$ ,  $k$  is temperature dependent rate constant (mol/m<sup>2</sup>/s),  $a_{H^+}^i$  is the activity of  $H^+$  and  $i$  is empirical reaction order accounting for catalysis by  $H^+$  in solution.  $f(\Delta G_r)$  is the rate dependence on Gibbs free energy of the reaction  $\Delta G_r$ , the chemical driving force of the reaction. Generally, transition State Theory (TST) for elementary reactions (Lasaga, 1981a; Lasaga, 1981b),

$$f(\Delta G) = 1 - \exp\left(\frac{\Delta G_r}{RT}\right) \quad (2)$$

The rate constant is dependent on temperature and a variety of different mechanisms (such as neutral, acid, and base mechanisms) (Eq. 3; Palandri and Kharaka, 2004). The rate constants, activation energies, and the relevant kinetic parameters are taken from Palandri and Kharaka (2004).

$$k = k_{nu}^{25} \exp\left[\frac{-E_{nu}}{R} \left(\frac{1}{T} - \frac{1}{298.15}\right)\right] + k_H^{25} \exp\left[\frac{-E_H}{R} \left(\frac{1}{T} - \frac{1}{298.15}\right)\right] a_H^n + k_{OH}^{25} \exp\left[\frac{-E_{OH}}{R} \left(\frac{1}{T} - \frac{1}{298.15}\right)\right] a_{OH}^n \quad (3)$$

where subscripts  $nu$ ,  $H$ , and  $OH$  indicate neutral, acid, and base mechanisms, respectively;  $k^{25}$  is the rate constant at 25 °C;  $E$  is the apparent activation energy;  $R$  is the universal gas constant;  $T$  is the absolute temperature;  $a$  is the activity of the aqueous species; and  $n$  is a power term.

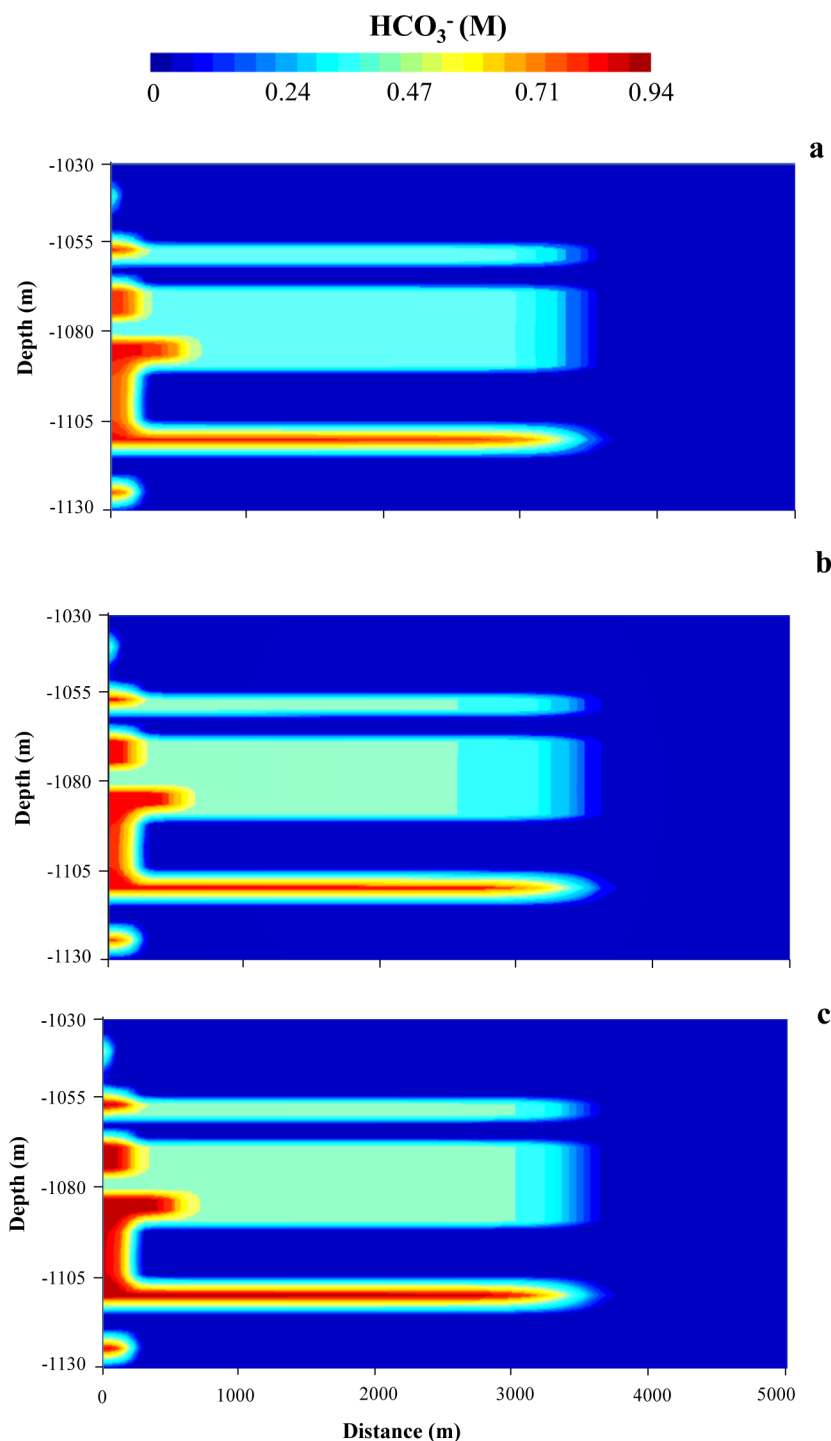
The modeling results allow us to evaluate the degree of two primary trapping mechanisms for dissolved  $CO_2$  storage:

- Solubility Trapping: Dissolution of  $CO_2$  in formation water leads to bicarbonate ( $HCO_3^-$ ) formation, preventing  $CO_2$  leakage.
- Mineral Trapping:  $CO_2$ -saturated water reacts with minerals (e.g., oligoclase and chlorite) to form stable mineral precipitates (calcite and ankerite).

## Modeling Results and Discussions

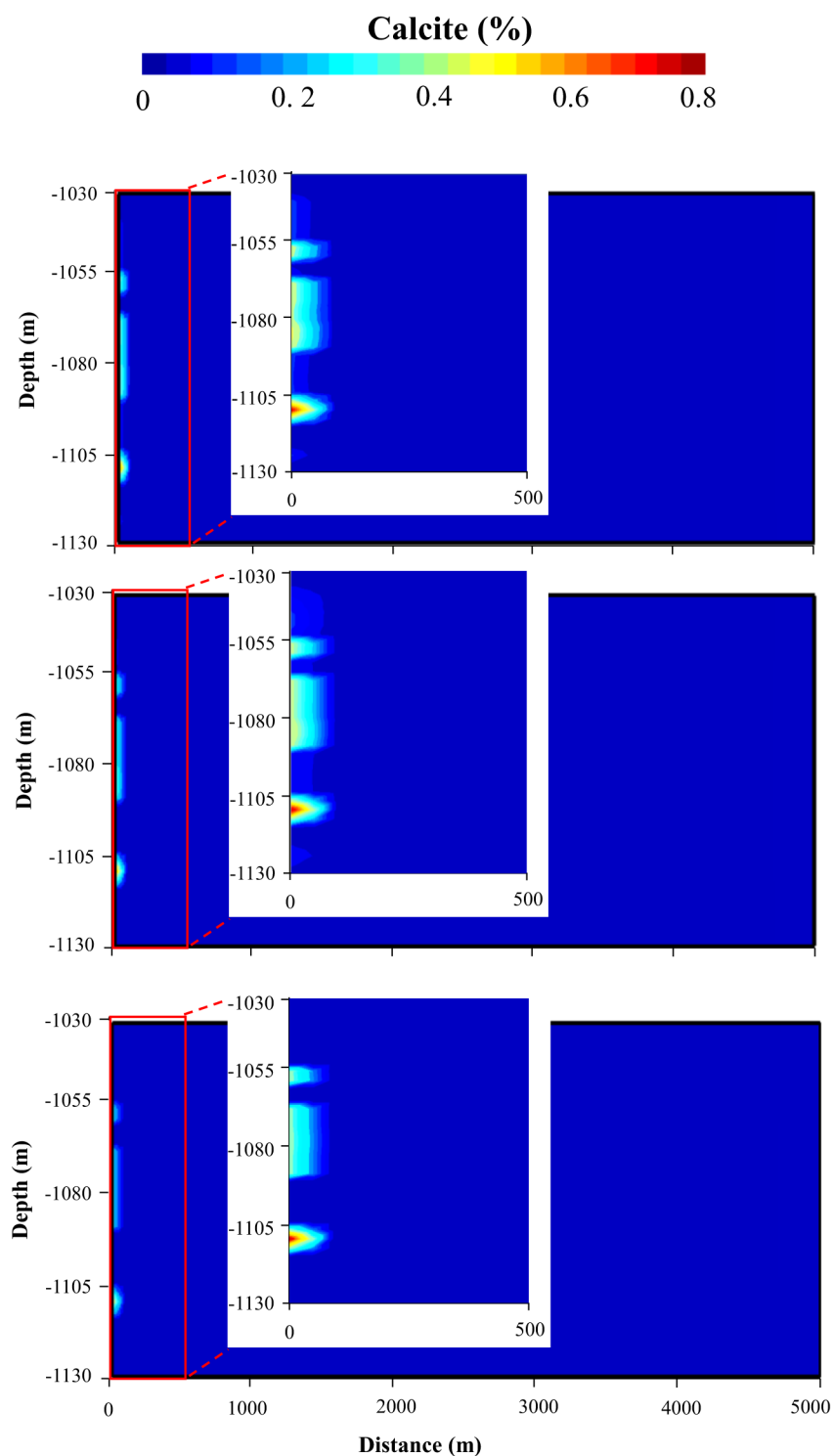
Figure 6 shows the spatial distribution of  $HCO_3^-$  plume at 10 year. Due to the reservoir heterogeneity, the migration of the  $HCO_3^-$  front is dependent on the permeability at specific locations that  $HCO_3^-$  front migrates further at the higher permeability zones. Figure 6 shows that  $HCO_3^-$  front at higher permeability zones migrates to ~ 3600 m away from the injection well at 10 year, while it only migrates ~ 100 m at the lower

permeability zones. The  $\text{HCO}_3^-$  concentrations of scenario 3 (up to 0.94 M) is much slightly than those in scenarios 1 and 2 ( $\sim 0.9$  and  $0.5$  M, respectively).



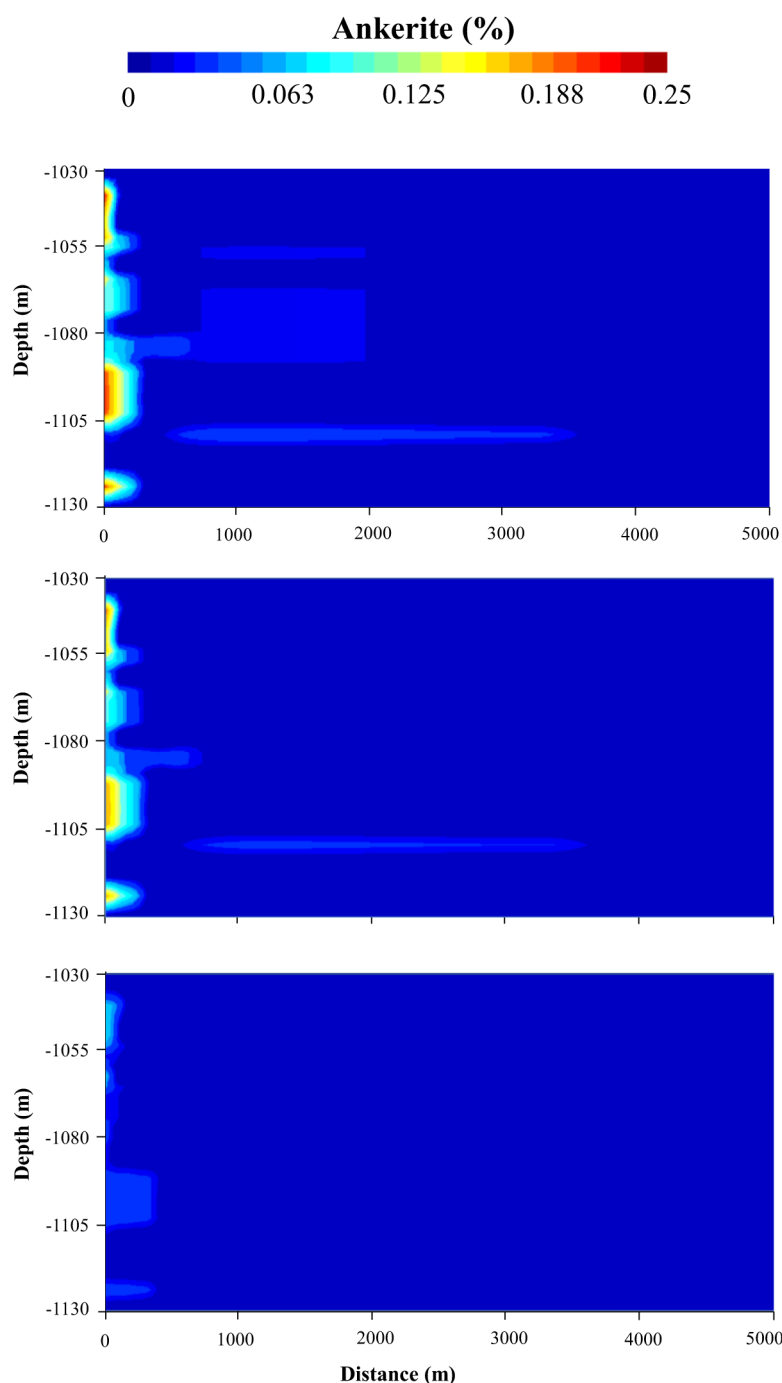
**Figure 6— $\text{HCO}_3^-$  (solubility trapping) after 100 yr's injection for different salinity formation water and injection water. (a) Scenario 1, (b) Scenario 2, (c) Scenario 3.**

Carbonate mineralization is mainly in the form of calcite ( $\text{CaCO}_3$ ) and ankerite ( $\text{CaMg}_{0.3}\text{Fe}_{0.7}(\text{CO}_3)_2$ ). The distribution of calcite is shown in Figure 7. The results show that calcite precipitation is focused near wellbore and restricted within 50 m radius of the injection well. The maximum abundance of calcite is  $< 1\%$  (0.8%).



**Figure 7—Calcite trapping after 100 yr's injection for different salinity formation water and injection water. (a) Scenario 1, (b) Scenario 2, (c) Scenario 3. The insets zoom in the first 500 m of the model domain to improve visualization of calcite distribution.**

Ankerite precipitation is found within 150 m radius of the injection well (Figure 8), which is wider than that of calcite. However, its maximum abundance is even less than that of calcite (cf. 0.25% vs. 0.8%). The abundance of ankerite shows a decreasing trend from Scenario 1 through Scenario 3, which is likely due to the lower  $\text{Mg}^{+2}$  and  $\text{Fe}^{+2}$  concentrations in the formation waters of Scenarios 2 and 3 associated with their lower salinity.



**Figure 8—Ankerite trapping after 100 yr's injection for different salinity formation water and injection water. (a) Scenario 1, (b) Scenario 2, (c) Scenario 3.**

The formation of calcite and ankerite near the injection well can establish a self-regulating feedback mechanism. In this process, fast-flowing fluids in preferred pathways increase the concentration of dissolved solids, which fosters the growth of mineral scales. As these scales develop, they reduce the porosity and permeability of the flow paths. This decrease in permeability then slows the flow rate, which in turn diminishes the rate of scale formation. Consequently, the flow in the preferred conduits is impeded, leading to a shift in flow behavior from focused to more uniform patterns. This change can enhance sweep efficiency, as highlighted by [Lu et al. \(2024\)](#).

The solubility of CO<sub>2</sub> in formation water was analyzed under different salinity conditions to determine the optimal storage depths and dissolution capacity. The results indicate that formation salinity and permeability play a crucial role in the spatial distribution of dissolved CO<sub>2</sub> over a 100-year period. (Figs. 7-8, Table 1):

- Dissolved CO<sub>2</sub> remained within the high-permeability zones, with minimal vertical migration.
- In low-TDS formation water and seawater injection, CO<sub>2</sub> dissolution was more evenly distributed, leading to enhanced solubility trapping.
- In high-TDS formation water injection, CO<sub>2</sub> dissolution was more localized near the injection well, with increased mineralization near the wellbore region.
- For mineral trapping, calcite precipitation is the dominant mineralization reaction. Calcite precipitation closed to the injection well and ankerite is more spread out.

**Table 1—Modeling cases and solubility and mineral trapping for different salinity formation water and injection water**

| Scenario | Salinity case (Formation water)* | Temp (°C) | Pressure (bars) | Injected CO <sub>2</sub> , mt/yr | Solubility trapping (%) | Mineral trapping (%) |
|----------|----------------------------------|-----------|-----------------|----------------------------------|-------------------------|----------------------|
| 1        | S1                               | 67        | 122             | 0.12                             | 95.07                   | 4.93                 |
| 2        | S2                               | 67        | 122             | 0.12                             | 95.62                   | 4.39                 |
| 3        | S3                               | 67        | 122             | 0.12                             | 98.06                   | 1.95                 |

\* S1: TDS 300000 ppm; S2: TDS 270000-290000 ppm; S3: TDS 70000-90000 ppm

## Conclusions

This study evaluated the feasibility of dissolved CO<sub>2</sub> storage in the shallow clastic reservoirs of an onshore basin of southern Red Sea as viable solution for carbon sequestration in shallow saline aquifers. The key findings of this study are as follows:

- Instantaneous solubility trapping is the dominant mechanism for CO<sub>2</sub> retention in low-salinity reservoirs, up to 98% (Table 1).
- Mineral trapping, through calcite and ankerite precipitation, prefers in high-TDS formations, but excessive mineralization near the injection well may negatively impact injectivity over time.

Overall, the study demonstrates that coastal shallow clastic reservoirs are well-suited for dissolved CO<sub>2</sub> storage, providing a safe and effective decarbonization solution for the region. The results indicate that this method can be successfully implemented in coastal basins along the Red Sea.

While this study confirms the feasibility of dissolved CO<sub>2</sub> storage, several areas require further investigation to optimize its practical implementation. 3D reservoir simulation is necessary to refine storage potential estimates and identify the most suitable injection "sweet spots" within the AOI. Field-scale pilot test is essential to validate the modeling results and assess real-world injection performance under varying reservoir conditions.

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